

PROJECT PLANNING PHASE

Date	30 September 2026
Team ID	SWTID-2026-3024
Project Name	AI-Augmented Backend Development AI BlogNest API
Maximum Marks	5 marks

1. PRODUCT BACKLOG, SPRINT SCHEDULE AND ESTIMATION

Sprint-1 – Authentication, Authorization and Security

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	User Registration & Authentication	USN-1	As a user, I can register for the application by providing the required details.	3	High	Padmashri – TL, Susmitha
Sprint-1	User Registration & Authentication	USN-2	As a registered user, I can log into the application using my email and password.	3	High	Padmashri – TL, Susmitha
Sprint-1	User Registration & Authentication	USN-3	As a user, my password should be securely protected using password hashing.	2	High	Padmashri – TL
Sprint-1	User Registration & Authentication	USN-4	As an authenticated user, I should receive a JWT token for accessing protected APIs.	3	High	Padmashri – TL
Sprint-1	Role-Based Access	USN-5	As an administrator, I can manage user roles and permissions.	3	High	Padmashri – TL, Subasri
Sprint-1	Security & Validation	USN-6	As a system, I should validate user input and reject invalid registration or login data.	2	High	Padmashri – TL, Sundari
Sprint-1	Security & Validation	USN-7	As a system, I should handle authentication errors and unauthorized requests appropriately.	2	High	Padmashri – TL, Sundari

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1 Total				18		

Sprint-1 Goal

To establish the secure foundation of AI BlogNest by implementing user registration, login, password protection, JWT authentication, role-based access control, and input validation.

Sprint-2 – Blog Management, Comments and Database

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Blog Management	USN-8	As an author, I can create a new blog post with the required content details.	3	High	Padmashri – TL, Subasri
Sprint-2	Blog Management	USN-9	As a user, I can view available blog posts.	2	High	Subasri, Padmashri – TL
Sprint-2	Blog Management	USN-10	As an authorized user, I can update an existing blog post.	3	High	Padmashri – TL, Subasri
Sprint-2	Blog Management	USN-11	As an authorized user, I can delete an existing blog post.	2	High	Subasri
Sprint-2	Blog Management	USN-12	As a user, I can organize blog content using categories and tags.	3	Medium	Padmashri – TL, Thejasri
Sprint-2	Comment Management	USN-13	As a user, I can add comments to blog posts.	2	Medium	Subasri, Sundari
Sprint-2	Comment Management	USN-14	As an authorized user, I can manage comments associated with blog posts.	2	Medium	Sundari, Padmashri – TL
Sprint-2	Database Management	USN-15	As a system, blog and user information should be stored and retrieved from MongoDB.	3	High	Padmashri – TL, Subasri

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2 Total				20		

Sprint-2 Goal

To implement the core blog management functionality and establish MongoDB database operations for users, blogs, comments, categories, and related information.

Sprint-3 – AI Integration and Content Assistance

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-3	AI Content Generation	USN-16	As a content creator, I can provide a topic or prompt to generate blog content using AI.	5	High	Padmashri – TL, Thejasri
Sprint-3	AI Content Generation	USN-17	As a user, I can receive AI-generated content through the REST API.	3	High	Padmashri – TL, Thejasri
Sprint-3	AI Content Assistance	USN-18	As a user, I can generate a concise summary of existing blog content using AI.	4	High	Padmashri – TL
Sprint-3	AI Content Assistance	USN-19	As a system, I should validate AI requests before sending them to the AI service.	2	Medium	Padmashri – TL, Sundari
Sprint-3	AI Integration	USN-20	As a system, I should handle AI service errors and return appropriate responses.	3	Medium	Padmashri – TL, Thejasri
Sprint-3	API Integration	USN-21	As a developer, I can integrate AI functionality with the existing backend API structure.	3	High	Padmashri – TL, Thejasri
Sprint-3 Total				20		

Sprint-3 Goal

To integrate Artificial Intelligence into the backend and provide AI-assisted blog generation, content summarization, validation, and error handling.

Sprint-4 – Testing, Documentation and Final Integration

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-4	API Testing	USN-22	As a developer, I can test authentication APIs using Postman.	2	High	Padmashri – TL, Sundari
Sprint-4	API Testing	USN-23	As a developer, I can test blog management APIs using valid and invalid requests.	3	High	Padmashri – TL, Sundari
Sprint-4	API Testing	USN-24	As a developer, I can test AI generation and summarization APIs.	3	High	Padmashri – TL, Thejasri
Sprint-4	Security Testing	USN-25	As a tester, I can verify unauthorized and invalid requests are handled correctly.	3	High	Padmashri – TL, Sundari
Sprint-4	Error Handling	USN-26	As a system, API errors should return meaningful and structured responses.	2	Medium	Padmashri – TL
Sprint-4	Documentation	USN-27	As a project team, we can prepare technical documentation for the completed system.	3	High	Padmashri – TL, Sundari
Sprint-4	Final Review	USN-28	As a project team, we can perform final testing and verify the major project requirements.	4	High	Padmashri – TL, Susmitha, Subasri, Thejasri, Sundari
Sprint-4 Total				20		

Sprint-4 Goal

To test the complete backend system, verify security and API functionality, resolve identified issues, prepare documentation, and complete the final project review.

2. COMPLETE PRODUCT BACKLOG SUMMARY

Sprint	Main Work	Story Points	Main Responsibility
Sprint-1	Authentication, Authorization & Security	18	Padmashri – TL, Susmitha, Sundari
Sprint-2	Blog Management, Comments & Database	20	Padmashri – TL, Subasri, Sundari, Thejasri
Sprint-3	AI Integration & Content Assistance	20	Padmashri – TL, Thejasri, Sundari
Sprint-4	Testing, Documentation & Final Review	20	Padmashri – TL, Sundari, Thejasri + Team
Total	Complete AI BlogNest Project	78	Entire Team

3. TEAM RESPONSIBILITY SUMMARY

Padmashri – Team Lead

Responsible for major project activities including overall project coordination, backend development, system architecture, API implementation, authentication and authorization integration, database integration, AI integration, API testing, debugging, final integration, and documentation.

Susmitha

Responsible for supporting user registration, login, authentication-related activities, and assisting with final project review and integration.

Subasri

Responsible for supporting blog management, CRUD operations, database-related activities, and blog content organization.

Thejasri

Responsible for supporting AI integration, AI content generation, AI-assisted features, categories/tags, and AI API testing.

Sundari

Responsible for supporting validation, security testing, comment management, API testing, documentation support, and verification of test results.

4. SPRINT SCHEDULE

Sprint	Sprint Total Story Points	Duration	Sprint Start Date	Sprint End Date	Planned Story Points Completed	Sprint Release Date
Sprint-1	18	6 Days	Week 1 - Day 1	Week 1 - Day 6	18	End of Sprint-1
Sprint-2	20	6 Days	Week 2 - Day 1	Week 2 - Day 6	20	End of Sprint-2
Sprint-3	20	6 Days	Week 3 - Day 1	Week 3 - Day 6	20	End of Sprint-3
Sprint-4	20	6 Days	Week 4 - Day 1	Week 4 - Day 6	20	End of Sprint-4
Total	78	24 Days			78	Final Release

5. SPRINT-1 PLANNING

Sprint Name: Authentication & Security

Sprint Goal:

Develop the secure user authentication and authorization foundation of AI BlogNest.

Major Tasks:

- User registration
- User login
- Password hashing
- JWT authentication
- Role-based access control
- Input validation
- Unauthorized access handling

Responsible Members:

Padmashri – TL, Susmitha, Sundari

Planned Story Points: 18

6. SPRINT-2 PLANNING

Sprint Name: Blog & Database Management

Sprint Goal:

Develop complete blog management functionality and establish database operations.

Major Tasks:

- Blog creation
- Blog retrieval
- Blog updating
- Blog deletion
- Categories and tags
- Comment management
- MongoDB integration
- Database operations

Responsible Members:

Padmashri – TL, Subasri, Sundari, Thejasri

Planned Story Points: 20

7. SPRINT-3 PLANNING

Sprint Name: AI Integration

Sprint Goal:

Integrate AI capabilities into the backend for intelligent content generation and summarization.

Major Tasks:

- AI service configuration
- AI blog generation
- AI content summarization
- AI request validation
- AI error handling
- AI API integration

Responsible Members:

Padmashri – TL, Thejasri, Sundari

Planned Story Points: 20

8. SPRINT-4 PLANNING

Sprint Name: Testing, Documentation & Final Review

Sprint Goal:

Verify the complete system through API testing and prepare the project for final submission and demonstration.

Major Tasks:

- Authentication API testing
- Blog API testing
- AI API testing
- Positive and negative testing
- Security testing
- Error handling verification
- Bug tracking
- Documentation
- Final project review

Responsible Members:

Padmashri – TL, Sundari, Thejasri, Susmitha, Subasri

Planned Story Points: 20

9. VELOCITY

Velocity represents the number of story points completed by the team during each sprint.

Sprint Story Points

Sprint-1	18
Sprint-2	20
Sprint-3	20
Sprint-4	20
Total	78

Average Velocity

Average Velocity = Total Story Points / Number of Sprints

Average Velocity = 78 / 4 = 19.5 story points per sprint

Therefore, the planned average team velocity for AI BlogNest is:

19.5 Story Points per Sprint

10. BURNDOWN CHART

A burndown chart represents the amount of remaining work against the development timeline. It helps the team monitor sprint progress and determine whether the planned work is being completed within the sprint duration.

Planned Sprint Burndown Data

Sprint	Starting Work	Remaining Work
Sprint-1	18	0
Sprint-2	20	0
Sprint-3	20	0
Sprint-4	20	0

Project Planning Conclusion

The AI BlogNest project was divided into four sprints based on functional requirements and development priorities. The planning process allocated **78 story points** across authentication, security, blog management, database operations, AI integration, testing, and documentation.

The sprint structure provides a systematic approach for developing the project while allowing the team to track progress, distribute responsibilities, estimate effort, and review completed work. As Team Lead, **Padmashri** handled a major share of the development, integration, coordination, testing, and documentation activities, while the other team members contributed to specific modules and supporting activities.